Building the Planck Length

a smallest ruler for space, found on a lattice

Year 11 entry point • Physics & Mathematics • 28 marks

Where this is going. By combining the three constants that govern relativity, the quantum world and gravity, you will build the only length that can be made from them,

$$\ell_P = \sqrt{\frac{h\,G}{c^3}},$$

a length of about 10^{-35} metres that may mark the smallest scale at which the idea of distance still means anything. To reach it we first talk about rates of change, ordinarily the territory of the calculus and then instead do everything on a *lattice*, using only subtraction, so that no notion of **continuity** or of **limits** is required.

Preamble: the SI base units we need

For this worksheet only three base units are in play, since no temperature appears; we track them as abstract *dimensions*.

Quantity	Base unit	Symbol	Dimension
length	metre	m	L
mass	kilogram	kg	M
time	second	s	T

Square brackets $[X]$ around the dimension are read "the dimension of X". Two quantities may be set equal only when their dimensions agree, a rule called *dimensional homogeneity*, and it is the whole engine of this investigation.

Working on a lattice, instead of a continuum

The mathematics of smoothly changing quantities, of gradients measured at a single instant, belongs to the calculus, which rests on ideas of continuity and of limits you have not yet met. We can step around all of that by refusing to look at a quantity at every instant and recording it only at a sequence of equally spaced moments, like photographs taken on a steady beat; the quantity then becomes a list of numbers, a lattice of samples, and everything we need follows from simple subtraction. This discrete way of working carries a deeper hint besides, since by the close you will meet a length so small that space itself may be woven from a lattice of roughly that spacing, in which case the discrete picture would be the truer one.

First differences, and a difference table

Given a list of values $f_0, f_1, f_2, \ldots$ recorded at equally spaced steps, the *first difference* at step n is the amount by which the list changes across that step,

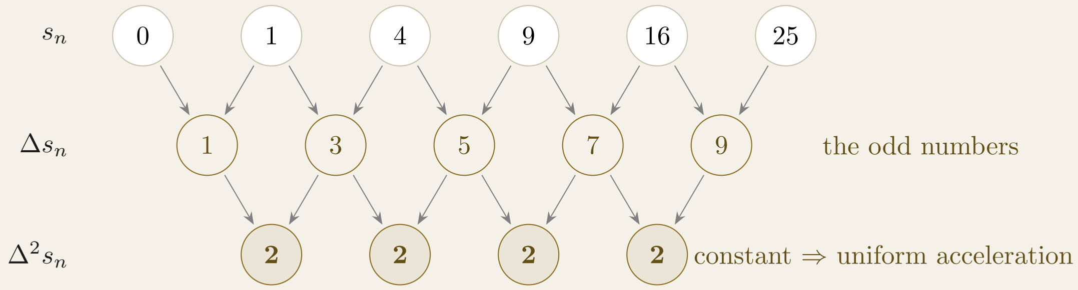
$$\Delta f_n = f_{n+1} - f_n.$$

When the values climb by the same amount each step they lie on a straight line and the first differences are all equal, which is the discrete counterpart of a constant gradient; when it is instead the first differences that climb by a constant amount, the *second differences*,

$$\Delta^2 f_n = \Delta f_{n+1} - \Delta f_n = f_{n+2} - 2f_{n+1} + f_n,$$

are constant, and this is the fingerprint of square-law growth. Writing the list, then the gaps between its entries, then the gaps between those gaps, builds a *difference table*.

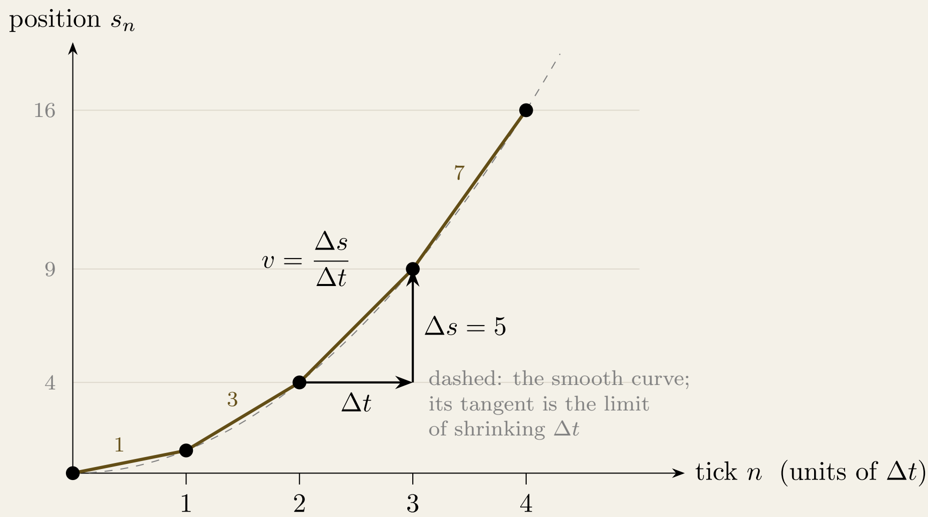
Figure 1: the difference cascade for $s_n = 0, 1, 4, 9, 16, 25$



Each entry in a lower row is the result of subtracting the two entries that point down into it, so the middle row holds the first differences and the bottom row the second differences.

1. (2 points) Using only the cascade, and without squaring, predict the next position s_6. State the next first difference you would write, and then the value of s_6.

Figure 2: reading a rate off the lattice



Each gold segment joins two successive samples; its steepness is the distance gained in one tick divided by that tick, the discrete speed $v = \Delta s/\Delta t$. The right-angled step shows this rise over run for the interval from $n = 2$ to $n = 3$.

2. (3 points) Read the figure to answer the following.
   (a) (1 point) State discrete speed $v = \Delta s/\Delta t$ for highlighted step from $n = 2$ to $n = 3$.

   (b) (2 points) The chord slopes read $1, 3, 5, 7$ along the four steps. State by how much the speed changes from one step to the next, and name the quantity this increase represents.

Rates without limits, and their dimensions

A rate of change is a change in one quantity divided by the change in another, and on the lattice we read it as a first difference of one list divided by the step in the other, $\Delta f/\Delta x$. The average speed across a time-step is the change in position over the tick, $v = \Delta s/\Delta t$, and the acceleration is the change in that speed over the next tick, $a = \Delta v/\Delta t$, which is the second difference of position spread over two ticks, $a = \Delta^2 s/(\Delta t)^2$. We never need the step to shrink to nothing; the continuum, and with it the calculus, would be what you reach by letting the spacing tend to zero, a road we deliberately leave untravelled.

Because subtracting two quantities of the same kind leaves a quantity of that same kind, a difference keeps the dimension of the thing differenced, $[\Delta f] = [f]$, and so a difference quotient carries the dimension of f divided by that of x, $[\Delta f/\Delta x] = [f]\,/\,[x]$. The consequence is the one that matters below: a discrete rate has exactly the same dimensions as the instantaneous rate the calculus would assign, so we can find the units of any rate, and of any constant defined through a rate, while working entirely on the lattice.

Section A | Reading motion off a lattice

A small ball is released from rest and its position s_n is recorded at equally spaced clock-ticks, giving the list below in some fixed unit of length.

n	0	1	2	3	4	5
s_n	0	1	4	9	16	25

1. (3 points) Build the difference table by completing the rows of first and second differences.

2. (2 points) State what the constant second difference tells you about the motion, and connect the first differences you found to the distances the ball covers in each successive tick.

3. (3 points) Treating the speed as $v = \Delta s/\Delta t$ and the acceleration as $a = \Delta v/\Delta t = \Delta^2 s/(\Delta t)^2$, where Δt is one tick, write down the dimensions $[v]$ and $[a]$.

Section B | The dimensions of the three constants

The dimensions of **force** and **energy** follow from the lattice rates you have just established, and are supplied here so that the constants can be read.

Foundations: force and energy

- **Force** obeys Newton's second law $F = ma$, force being proportional to acceleration with the mass as the constant of proportionality, so $[F] = [m]\,[a] = \mathsf{M\,L\,T}^{-2}$ (the newton).
- **Energy** can be read from kinetic energy $E_{\text{kin}} = \frac{1}{2}mv^2$, with m the mass and v the speed, so $[E] = [m]\,[v]^2 = \mathsf{M}\,(\mathsf{L\,T}^{-1})^2 = \mathsf{M\,L^2\,T}^{-2}$ (the joule). The $\frac{1}{2}$ is a pure number.

4. (1 point) **The speed of light, c.** Light in a vacuum covers equal distances in equal ticks, so as a difference quotient $c = \Delta x/\Delta t$, where Δx is the distance crossed (in m) and Δt the time taken (in s). Write the dimension $[c]$.

5. (2 points) **The Planck constant, h.** A photon's energy is proportional to the frequency of its light, $E = hf$, where E is the photon energy (in J) and f is the frequency, a count of oscillations each second, so $[f] = \mathsf{T}^{-1}$. Rearrange for h and write its dimension $[h]$.

6. (3 points) **The gravitational constant, G.** The attraction between two masses is proportional to each mass, and falls as $1/r^2$ because a fixed influence spreads over the surface of a sphere of area $4\pi r^2$ as it streams outward; hence $F = G\,m_1 m_2/r^2$, where F is the force (in N), m_1, m_2 are the masses (in kg) and r is their separation (in m). Rearrange for G and write its dimension $[G]$.

Section C | Forming the Planck length

We look for a length, ℓ_P built from c, h and G alone, written as a product of powers with a pure number κ in front:

$$\ell_P \;=\; \kappa\, c^{\alpha}\, h^{\beta}\, G^{\gamma}.$$

7. (4 points) Using the dimensions $[c] = \mathsf{L}\,\mathsf{T}^{-1}$, $[h] = \mathsf{M}\,\mathsf{L}^{2}\,\mathsf{T}^{-1}$ and $[G] = \mathsf{M}^{-1}\,\mathsf{L}^{3}\,\mathsf{T}^{-2}$, and requiring the right-hand side to have dimension $[\ell_P] = \mathsf{L}$, write down the three equations obtained by matching M, L and T.

8. (3 points) Solve the three equations for α, β, γ, and hence write ℓ_P in surd form.

9. (1 point) Unlike many such problems, this one has a single solution for the powers with no free choice left over. State briefly why.

Section D | Its size, and what it might mean

10. (2 points) Evaluate $\ell_P = \sqrt{\hbar G/c^3}$ taking $\kappa = 1$ with the reduced Planck constant $\hbar = h/2\pi$, using $\hbar = 1.05 \times 10^{-34}\,\mathrm{J\,s}$, $G = 6.67 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}$ and $c = 3.00 \times 10^{8}\,\mathrm{m\,s^{-1}}$.

11. (1 point) Had we built the length from h rather than from $\hbar = h/2\pi$, the value would have come out about $\sqrt{2\pi}$ times larger. Explain why dimensional analysis alone could never have told us which of these to use.

12. (2 points) In a companion investigation on black-hole entropy you will meet five unknown powers but only four base dimensions, so the units do not pin everything down and a physical assumption is required. Explain, by contrast, why the Planck length needed no such assumption.

13. (1 point) Squaring the Planck length gives the *Planck area* $\ell_P^2 = \hbar G/c^3$, a tile of about $2.6 \times 10^{-70}\,\mathrm{m^2}$. Write an expression for the number of such tiles needed to cover a horizon of area A, the quantity that the entropy worksheet will show is being counted.